

# External Evidence and Language-Model Forecasts: A Blinded Comparison with Prediction-Market Prices

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## Abstract

We examine whether access to current primary-source evidence improves a language model’s agreement with prediction-market prices. Using the Vorhersage forecasting workbench, we compare question-only and evidence-supplied forecasts for six unresolved Kalshi contracts across six subject areas. The model, instructions, and response schema are held fixed, with one independent call per contract and condition. Opening market quotes are withheld until all forecasts are recorded and sealed; Kalshi is excluded as a source of research evidence. Mean absolute deviation from the opening bid–ask midpoint is 33.17 percentage points in the question-only condition and 8.58 points in the research condition. Research is associated with a smaller deviation in five cases and an unchanged deviation in one. A documented entity-identification error persists in the researched Netflix forecast. The small purposive sample, single draw per condition, and use of prices rather than resolved outcomes limit interpretation. The study provides a reproducible pilot of evidence ablation and market agreement, rather than evidence of calibrated forecasting accuracy or a general advantage from a particular forecasting workflow.

## 1 Introduction

A forecasting system must translate evidence about an event into a numerical probability. A useful preliminary question is whether supplying current, relevant facts changes those probabilities in a direction consistent with an external benchmark. This study addresses that question by comparing two information conditions on the same set of unresolved prediction-market contracts. One condition supplies only the event definition; the other adds a frozen packet of primary-source evidence. Both use the same language model and forecasting instructions.

The comparison uses contemporaneous market midpoints as reference values. These values provide an immediate benchmark before events resolve, but they are neither known event probabilities nor observed outcomes. We therefore measure *market agreement*, defined as proximity to a recorded quote, and reserve claims about forecasting accuracy and calibration for evaluation against subsequent outcomes. The design also requires that neither target prices nor market commentary enter the forecasting inputs.

Vorhersage provides the registration, evidence storage, forecast validation, and separation of hidden targets from model inputs. The present experiment uses a minimal direct-judgment method to isolate the information contrast as far as possible. It does not compare the package’s full structured workflow with an unstructured alternative. The dated protocol, registration, and result artifacts provide the empirical record for the analysis [1, 2].

## 2 Experimental design

### 2.1 Contract selection and target measurement

The sample consists of six contracts selected to span weather, employment, monetary policy, space technology, entertainment, and sports. Candidate contracts and ordered substitutes were recorded before order-book retrieval. Previously revealed cases were excluded. Eligibility required an active, unresolved binary contract; an uncrossed, non-extreme two-sided book; at least one contract at each best quote; and a bid–ask spread no greater than ten cents. Two initial candidates failed these requirements and were replaced by predefined substitutes. The researcher received eligibility flags without seeing accepted or rejected quotes. This procedure limits direct selection on observed prices, although quote eligibility still conditions the sample. The cohort was selected purposively rather than sampled randomly.

Opening books were captured on September 18, 2026, between 11:06:23 and 11:06:26 UTC (7:06 a.m. EDT), before external research. The reference value is the arithmetic midpoint of the best YES bid and ask. Equal weight is assigned to each contract, irrespective of quoted size, topic, or forecast horizon. Table 1 identifies the events and recorded probabilities.

### 2.2 Information conditions and forecasting procedure

An experimental *arm* in Vorhersage binds a versioned method, model configuration, and data assignment. Both arms used Google’s *gemini-3.1-pro-preview* with identical instructions and output schemas. The question-only arm received the precise event and settlement conditions. The research arm received those inputs plus a frozen packet containing source excerpts, dates, extracted facts, and limitations. An external coding agent selected and collected the evidence from official sources; the forecasting model itself had no browsing tools or local filesystem access. Across the six packets, nine source records were supplied. Appendix B describes their contents and omissions.

Each contract received one fresh model call per arm, producing twelve forecasts. Both arms were generated after evidence collection, in separate contexts. Consequently, the question-only estimate is an information ablation, not a chronological pre-research forecast. The common task requested a probability, a concise rationale, reasons the estimate could be too high or too low, and limitations. The method had an assessment stage followed by deterministic issuance, with no model review or adjudication. Full settings and execution details appear in Appendix A.

### 2.3 Blinding and provenance

Kalshi supplied contract definitions and evaluator targets, but no Kalshi material was used as research evidence. The package exposed only allowlisted contract-definition fields to the researcher. Opening order books were saved in a separate evaluator database; the research project stored commitments to the hidden targets. Research was restricted to approved primary-source domains. URL and redirect checks, together with lexical screening of returned text, were applied before display to exclude prediction-market and betting-odds material. Forecasting prompts contained only the assigned question, common instructions, and permitted evidence.

All twelve forecasts were sealed at 11:25:10 UTC. Targets were revealed between 11:40:25 and 11:40:27 UTC, after the study’s reveal routine verified an unchanged forecast seal and a terminal attempt for every registered trial. The subsequent audit checked target commitments, evidence and forecast hashes, and recorded timing [3]. The design provides auditable procedural blinding

rather than independently enforced isolation: the coordinator could technically access the evaluator database. Lexical filters cannot exclude every indirect reference, and model pretraining exposure is unobserved. The information supplied to the hosted calls is inspectable, but these checks cannot establish the absence of every possible information channel.

## 2.4 Evaluation measures

Let  $i = 1, \dots, n$  index contracts, with  $n = 6$ , and let  $k \in \{Q, R\}$  denote the question-only and research conditions. Write  $b_i$  and  $a_i$  for the opening YES bid and ask,  $q_i = (b_i + a_i)/2$  for their midpoint, and  $p_{ik}$  for the model forecast. The primary descriptive measure is mean absolute deviation from the midpoint, expressed in percentage points:

$$\text{MAE}_k = \frac{100}{n} \sum_{i=1}^n |p_{ik} - q_i|. \quad (1)$$

We also report root mean squared deviation and mean distance outside the opening bid–ask interval:

$$\text{RMSE}_k = 100 \sqrt{\frac{1}{n} \sum_{i=1}^n (p_{ik} - q_i)^2}, \quad (2)$$

$$\text{Outside}_k = \frac{100}{n} \sum_{i=1}^n \max\{b_i - p_{ik}, p_{ik} - a_i, 0\}. \quad (3)$$

The paired improvement for contract  $i$  is  $\Delta_i = 100(|p_{iQ} - q_i| - |p_{iR} - q_i|)$ ; positive values indicate closer agreement in the research arm. A constant forecast of 0.5 provides an additional reference. No outcome-based Brier score is computed. Neither the bid–ask interval nor the variation across these six purposively selected contracts is treated as a confidence interval for predictive accuracy.

## 3 Results

### 3.1 Aggregate and contract-level agreement

Mean absolute deviation is 33.17 percentage points for question-only forecasts and 8.58 points for research forecasts, a paired mean reduction of 24.58 points. Research yields closer agreement in five contracts and identical agreement in the remaining contract (Figure 1). No contract was excluded after forecasting. All twelve responses passed validation on their first attempt; none was repaired or rerun.

The constant-50% reference has a mean absolute deviation of 27.75 points, falling between the two model conditions. Research also has lower root mean squared deviation (11.40 versus 40.83 points) and lower mean distance outside the opening spread (7.67 versus 32.25 points). Table 2 reports all three measures on the same six-contract cohort. Follow-up quotes collected at reveal have unchanged midpoints for every contract, so the midpoint-based results are identical using those later observations. Netflix’s spread widens from three to seven cents.



**Figure 1. Forecast probabilities and opening Kalshi midpoints by question.** Each panel connects the question-only forecast on the left to the outside-research forecast on the right. A labeled horizontal dashed line marks the opening Kalshi midpoint. The vertical distance from each forecast to this line is its absolute market deviation. Boxed slope labels report the reduction in that deviation in percentage points and, in parentheses, as a percentage of the question-only deviation:  $100(d_Q - d_R)/d_Q$ , where  $d_k = |p_k - q|$ . Positive values indicate improvement. Endpoint labels report probabilities; all panels share a 0–100% scale. The two forecasts are separate information conditions, not chronological updates.

**Table 1.** Contract definitions and probabilities.

| YES event                             | Opening market (%) |          | Forecast (%)  |          |
|---------------------------------------|--------------------|----------|---------------|----------|
|                                       | Bid-ask            | Midpoint | Question only | Research |
| LAX high 74–75°F (Sep. 18)            | 3–4                | 3.5      | 12            | 5        |
| US unemployment > 4.4% (Sep.)         | 1–2                | 1.5      | 50            | 3        |
| Fed holds rates (Oct. 28)             | 48–49              | 48.5     | 55            | 55       |
| Neutron launches before 2027          | 4–8                | 6        | 65            | 20       |
| Netflix film ranks first <sup>a</sup> | 61–64              | 62.5     | 0.5           | 40       |
| Milwaukee beats Baltimore (Sep. 20)   | 63–64              | 63.5     | 49            | 58       |

Notes: All dates are in 2026. <sup>a</sup>*Why Did I Get Married Again?* ranks first in the US Netflix film chart published September 22 for the week ending September 20. The contract headline refers to September 21; settlement rules define the event.

**Table 2.** Deviation from the opening market benchmark.

| Measure (percentage points)  | Question only | Research | Constant 50% |
|------------------------------|---------------|----------|--------------|
| Mean absolute gap            | 33.17         | 8.58     | 27.75        |
| Root mean squared gap        | 40.83         | 11.40    | 33.64        |
| Mean distance outside spread | 32.25         | 7.67     | 26.83        |

Notes: Equal-weight averages over all six contracts. The benchmark is a quoted market price, not a resolved event outcome. The constant-50% condition is computed mechanically and requires no additional model calls.

### 3.2 Evidence use and observed failure modes

The largest reductions in absolute deviation occur for unemployment (47 points), Neutron (45 points), and Netflix (39.5 points). For unemployment, the BLS reading of 4.1% provides a current anchor: the contract’s strict threshold requires a published rate of at least 4.5%. The forecast falls from 50% to 3%. For Neutron, the company update concerns Q4 delivery to the launch pad, rather than a guaranteed launch that quarter. The forecast falls from 65% to 20%, but remains fourteen points above the market midpoint.

The Netflix case illustrates a limitation of both source selection and model interpretation. Both responses reason as though the target were an older catalog film. Official discovery identified a September 2026 release, but the packet preserved the previous week’s number-one ranking without the release date. This omission and the entity-identification error were recorded at 11:25:43 UTC, before target reveal [3]. The original forecasts remain in the analysis: research raises the estimate from 0.5% to 40%, compared with a midpoint of 62.5%.

Smaller reductions occur for Milwaukee baseball (nine points) and LAX weather (seven points). Team records and run differentials favor Milwaukee, but the packet lacks starting pitchers, lineups, and bullpen context. For LAX, an NWS forecast high of 79°F lies outside the 74–75°F event interval, reducing the forecast from 12% to 5%. The probability assigned to that interval is not supported by an estimated forecast-error distribution. Finally, both Fed forecasts are 55%, despite the researched explanation incorporating the latest policy decision and year-end projections. Equal probabilities therefore do not imply equal factual grounding.

## 4 Discussion

The observed pattern is consistent with current evidence improving market agreement in this cohort. Its interpretation is narrower than a general claim about research quality. The question-only condition lacks fresh facts that are directly relevant to several events. The comparison therefore does not distinguish a minimal factual update from deeper, adaptive research. Nor does it estimate the contribution of review, scenario modeling, ensembles, or other features of the package. Source collection was performed by an external coding agent, so the results do not measure autonomous search by the forecasting model.

Several additional limitations matter. First, six purposively selected contracts do not support a population-level performance estimate. Employment and monetary policy are related, and contract horizons differ substantially. Second, one draw per condition leaves model sampling variability unmeasured; the observed paired differences combine information-condition differences with stochastic generation. We report no significance test or inferential interval. Third, prices may reflect liquidity and trading frictions as well as beliefs. Neutron’s opening best bid has a size of only five contracts, which limits its interpretation as a precise reference value. Fourth, research follows the opening snapshots. Unchanged follow-up midpoints reduce concern about observed price movement during the run but do not establish identical information sets. Finally, procedural input separation does not certify the absence of unobserved leakage or pretraining exposure.

The pilot also identifies a concrete design improvement: both information conditions in a subsequent research-depth comparison should include verified entity identities and relevant release or publication dates. A fresh, unrevealed cohort could then compare a small current-facts packet with additional targeted research, using repeated forecasts and registered handling of failed submissions. Larger cohorts would permit paired uncertainty estimates that account for related event groups. Evaluation against resolved outcomes would be required to determine whether greater market agreement translates into better forecasts.

## References

- [1] Study protocol and experimental registration (2026). Vorhersage blinded Kalshi pilot, September 18. `protocol.json`, `run/selection.json`, and `run/registration.json`.
- [2] Forecasts and market comparison (2026). Vorhersage blinded Kalshi pilot, September 18. `run/comparison.json`, `run/raw-records.json`, and `run/revealed-targets/`.
- [3] Integrity audit and pre-reveal diagnostics (2026). Vorhersage blinded Kalshi pilot, September 18. `run/audit.json` and `run/pre-reveal-diagnostics.json`.

## A Package architecture and execution

Vorhersage 0.3.0 is a Python forecasting workbench with a command-line interface, requiring Python 3.11 or later and no declared runtime dependencies. It preserves questions, evidence, model inputs, forecasts, and evaluation records. The external agent chooses sources and makes judgments; deterministic package code checks schemas, bookkeeping, and declared numerical calculations. Stored provenance makes a claim inspectable but does not establish that it is true.

### A.1 Components and responsibilities

| Component                                       | Role in this study                                                                                                                                                                                |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>store.py</code> , <code>schemas.py</code> | SQLite transactions, versioned questions, immutable artifact records, content hashes, and validated submissions preserve what each trial received and produced.                                   |
| <code>experiments.py</code>                     | Registers a versioned method and separate arms binding method, model configuration, and question-specific evidence. Freezes the cohort, cutoffs, repetitions, and trial order.                    |
| <code>evidence.py</code>                        | Stores source captures, URLs, timestamps, excerpts, facts, and limitations in packets linked to the assigned arm.                                                                                 |
| <code>workflow.py</code>                        | Advances each trial through assessment and issuance; validates the submitted probability and rationale, then records the forecast.                                                                |
| <code>market_data.py</code>                     | Separates price-free contract definitions from private quote snapshots; derives the YES ask from the complementary NO bid.                                                                        |
| <code>workbench.py</code>                       | Keeps targets in a separate evaluator project, stores commitments, and requires completion before reveal.                                                                                         |
| Study scripts                                   | <code>study.py</code> coordinates selection, source filtering, registration, import, forecast sealing, reveal, and audit. <code>build_evidence.py</code> records the source-to-packet extraction. |

### A.2 Arm specification and model configuration

The shared method, `outside-source-direct-judgment`, uses only `assessment` and `issue`. Assessment asks for a probability, concise rationale, reasons it could be too high or too low, and limitations. Issuance records the result; it is not another model call. The question-only arm assigns no research packet, while the outside-research arm assigns its question’s frozen packet. Changing a registered method or arm requires a new version.

Both arms used temperature 0.5, topK 40, topP 1, `maxOutputTokens` 8192, and `thinking_budget` 2048. The model was called through an external EDSL transport with tools disabled. This transport and the report’s plotting dependencies are separate from the dependency-free core. Prompts, provider records, raw answers, accepted submissions, and issued forecasts were all retained. Model identity and settings are those recorded by the transport, not an attestation about the provider’s internal implementation.

### A.3 Scope of the evaluated implementation

The package also supports scenario mixtures, grouped likelihood-ratio updates, timeline models, reference-class priors, review stages, ensembles, and scoring against resolved outcomes. Those

features were not used to produce these twelve forecasts. Exa and Firecrawl adapters can preserve research-provider receipts, but this run used browser tools and direct official HTTP requests because their API keys were not configured. Thus the result concerns a frozen-evidence, direct-judgment comparison, not the performance of every available workflow.

## A.4 Integrity checks and recorded costs

The live audit verified all twelve issued forecasts and six opening-target commitments, with 82 artifacts passing database integrity checks. Fourteen focused tests covered source restrictions, quoted-odds rejection, premature or tampered reveal prevention, and metric arithmetic. All twelve model responses were accepted on their first attempt. Summed provider-reported model cost was \$0.16835. Research, browser, and coordinator dollar costs are unknown, not zero; this number is not total system cost. The ledger records fourteen direct HTTP attempts (seven successful), six browser searches, and six browser page-read operations across five tool calls.

# B Evidence and reproducibility

## B.1 The six frozen evidence packets

The packets contain nine source records. The following links identify the original sources; the dated captures in the run are authoritative for what the model received, since live pages may subsequently change.

*Weather.* [NWS point forecast](#) and the [LAX climate report](#): forecast high 79°F, previous high 78°F, and normal high 76°F. The packet does not estimate forecast errors and uses NWS as a proxy for the contract’s commercial settlement source.

*Employment.* [BLS employment report, September 4](#): August and July unemployment 4.1%, June 4.2%, and August payroll growth 162,000. The September report was scheduled for October 2. The contract’s strict threshold requires a published rate of at least 4.5%.

*Monetary policy.* [September 16 FOMC statement](#) and [economic projections](#): a 25-basis-point hike to 3.75–4.00%, unanimous 12–0 vote, and a median year-end rate projection of 4.1%. Year-end projections do not identify the meeting at which a subsequent change occurs.

*Space technology.* [Rocket Lab second-quarter update, August 10](#): first-flight assembly and testing, with first-stage tank delivery to the pad aligned to Q4. The issuer’s forward-looking update lacks a complete remaining test and launch-readiness schedule.

*Entertainment.* [Netflix US film chart](#): the target title ranked first for September 7–13. Its [September 9 release date](#) was found during discovery but omitted from the model’s packet. The error was logged in `run/pre-reveal-diagnostics.json` at 11:25:43 UTC.

*Sports.* [MLB schedule](#) and [standings](#): Milwaukee at Baltimore on September 20; records of 95–58 and 75–78, with run differentials of +197 and –26. No usable starting-pitcher, lineup, injury, or bullpen context entered the packet.

## B.2 Research artifacts and reproducibility

Paths are relative to `examples/kalshi_blind_20260918/`. The protocol and candidate order preceded price inspection; experiment registration followed evidence collection at 11:19:02 UTC.

| Record                                                                   | What it establishes                                          |
|--------------------------------------------------------------------------|--------------------------------------------------------------|
| <code>protocol.json</code> ; <code>run/selection.json</code>             | Planned comparison, source rules, and ordered candidates.    |
| <code>run/registration.json</code> ;<br><code>run/*-arm.json</code>      | Frozen questions, method/model/data assignments, and trials. |
| <code>run/packets/</code> ; <code>run/tasks/</code>                      | Exact evidence and prompts available to each call.           |
| <code>run/raw-records.json</code> ;<br><code>run/attempts.json</code>    | Responses, attempt outcomes, and model costs.                |
| <code>run/sealed-forecasts.json</code> ;<br><code>run/audit.json</code>  | Forecast seal, hash checks, and timing.                      |
| <code>run/comparison.json</code> ;<br><code>run/revealed-targets/</code> | Numerical results and original revealed target receipts.     |

`plot_performance.py` recomputes all three aggregate metrics for all three conditions and verifies them against the frozen comparison before generating the table rows and calling `plot_performance.R` for the faceted slope graph. The R script also verifies the metrics and renders the figure using `ggplot2`. With Python, R (`ggplot2` 3.5.0 or later and `jsonlite`), and LaTeX available, run from this study directory:

```
python plot_performance.py
latexmk -pdf -interaction=nonstopmode -halt-on-error report.tex
```

Rebuilding makes no model or market calls. A new blind experiment requires unrevealed contracts and a separate directory preserving the registration, sealing, and reveal order.